BACSE324 · EMBEDDED SYSTEMS ARCHITECTURE · FALL 2026 · FINAL REPORT

[Project title]

Final report · concept to realisation · due end of October · 15–20 pages excluding front matter and appendices

Scope tier	
Submission date	
Repository link	
Demo video link	

TEAM

Member	Reg. no.	Module owned	Sections written

DECLARATION

We declare that this report and the work it describes are our own, and that every external source, library and tool is acknowledged in the references.

Member	Signature	Date

Abstract

150 words. Problem, approach, the key result as a number against its target, and the main limitation.

Contents

Right-click → Update Field to build the contents.

1 Problem and requirements

The report is built on this table. Give every requirement an ID, a measurable target and, by the end, the section where it is verified. A requirement you cannot measure is a wish — rewrite it.

1.1 Problem statement

1.2 Requirements

ID	Requirement	Measurable target	Priority (must / should)	Verified in
R-F1				
R-F2				
R-F3				
R-N1				
R-N2				

F = functional (what it does). N = non-functional (latency, accuracy, power, cost, size).

1.3 Constraints

Budget, available hardware, power source, time, team size.

2 Alternatives considered

For each major decision — microcontroller, key sensor or actuator, communication link, software architecture — compare at least two options against weighted criteria taken from your requirements. Show the matrix, then state the choice.

2.1 [Decision, e.g. microcontroller]

Criterion (requirement ID)	Weight	Option A	Option B	Option C

Chosen: · **Because:**

2.2 [Decision]

Criterion (requirement ID)	Weight	Option A	Option B	Option C

Chosen: · Because:

3 Architecture and hardware

3.1 System architecture

Insert figure here — replace this box

Figure — System block diagram, final version.

3.2 Schematic and pin map

Insert figure here — replace this box

Figure — Circuit schematic.

MCU pin	Function	Connected to	Notes (pull-up, level shift, interrupt)

MCU pin	Function	Connected to	Notes (pull-up, level shift, interrupt)

3.3 Power budget

Component	Supply voltage	Active current	Sleep current	Duty cycle	Average current

3.4 Build

Photographs of the assembled system, labelled.

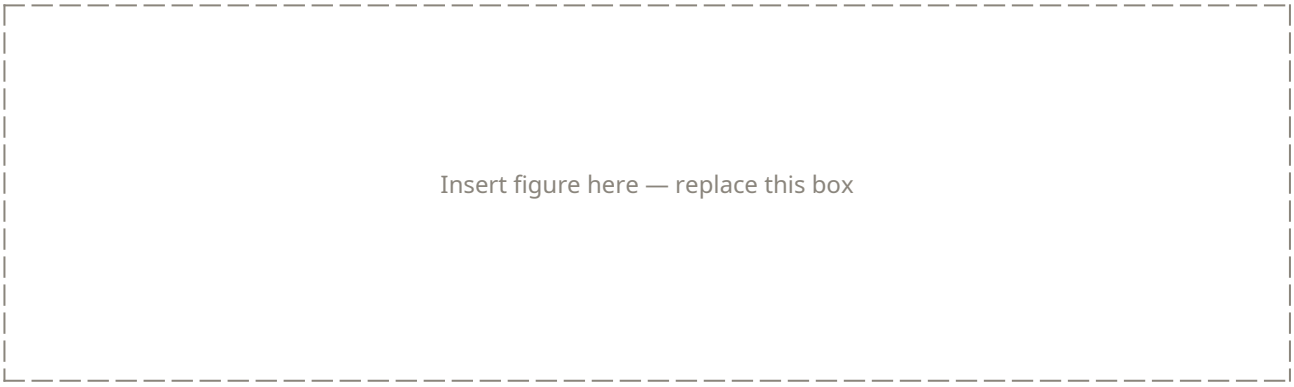


Figure — Assembled hardware.

4 Design models

Carry the CDFG, FSM and schedule forward from Review 2, revised to match what you built. Put the Review 2 versions in an appendix if they changed substantially.

4.1 CDFG and critical path

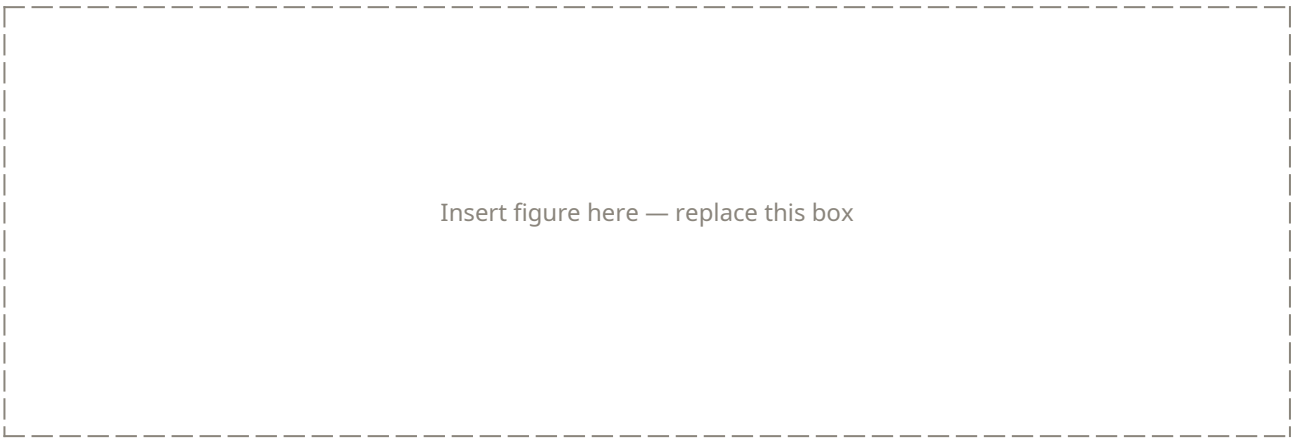


Figure — CDFG of the critical function, final.

4.2 System FSM



Figure — System FSM, final.

4.3 Task schedule



Figure — Schedule timeline, final.

4.4 Changes since Review 2

Model element	Review 2 version	As built	Why it changed

Model element	Review 2 version	As built	Why it changed

5 Firmware structure

5.1 Layers and modules

Driver, service and application layers, or your own decomposition. Show which module owns which peripheral.



Figure — Firmware module diagram.

5.2 Implementation of the models

How the FSM is coded (switch-case, state table, function pointers), how the schedule is realised (timer ISR, cooperative loop, RTOS), and where faults are caught.

5.3 Model-to-code traceability

Model element	File	Function	Commit / line

6 Optimisation

Pick at least one deliberate optimisation — memory, latency, power or code size. Report the baseline, the change, the mechanism and the result. Take RAM and flash from the linker map or build output, not from estimates.

Metric	Baseline	Optimised	Change	Mechanism	Requirement
Flash (bytes)					
RAM (bytes)					
Critical-path latency					
Average current					

Trade-off accepted:

7 Test plan and measured results

Every requirement from 1.2 needs at least one test. State the method and equipment, repeat each measurement, and report spread — not a single best run. Include at least one fault-injection test.

Test ID	Requirement	Method and equipment	Expected	Measured (n, mean, spread)	Pass / fail
T1	R-F1				
T2					
T3					
T4	R-N1				
T5 (fault)					

7.1 Timing measurements

Measured WCET and jitter for each task against the Review 2 schedule. A GPIO toggle captured on a scope or logic analyser is sufficient.

Task	WCET (Review 2)	WCET measured	Jitter	Deadline met?

Insert figure here — replace this box

Figure — Logic analyser or scope capture.

7.2 Discussion

What the results show, where they fall short of the targets, and why.

8 Design-decision log

Keep this running from the start of the project. Every entry names the alternatives rejected and the evidence used. Markers read this to judge justification.

ID	Date	Decision	Alternatives rejected	Reason and evidence	Consequence
D1					
D2					
D3					
D4					
D5					

9 Bill of materials and cost

Item	Part number	Qty	Unit cost (₹)	Source	Total (₹)
Total					

10 Individual contribution

Name the module each member owned and point to evidence: commits, files, measurements. Each member must be able to defend their module at the presentation.

Member	Module owned	Evidence (files, commits)	Report sections	Share %

11 Limitations and future work

What did not work or fell short of a target, what you would do differently, and the next engineering step. Stated failure scores higher than silence.

References

IEEE style. Datasheets, application notes, libraries and any tools used.

[1]

Appendices

Not counted in the page limit: full source listing or repository link, full schematics, raw measurement data, Review 2 models if they changed.

A Source code

B Raw data

Submission checklist

Item	✔
15–20 pages excluding front matter and appendices; guidance blocks deleted	
Every requirement has an ID, a measurable target and a test	
At least two alternatives compared for each major decision	

Item	✓
Models revised to match the build; changes since Review 2 listed	
Traceability table from model elements to code	
Optimisation with measured before and after	
Repeated measurements with spread; one fault-injection test	
Decision log, BOM and contribution table complete	